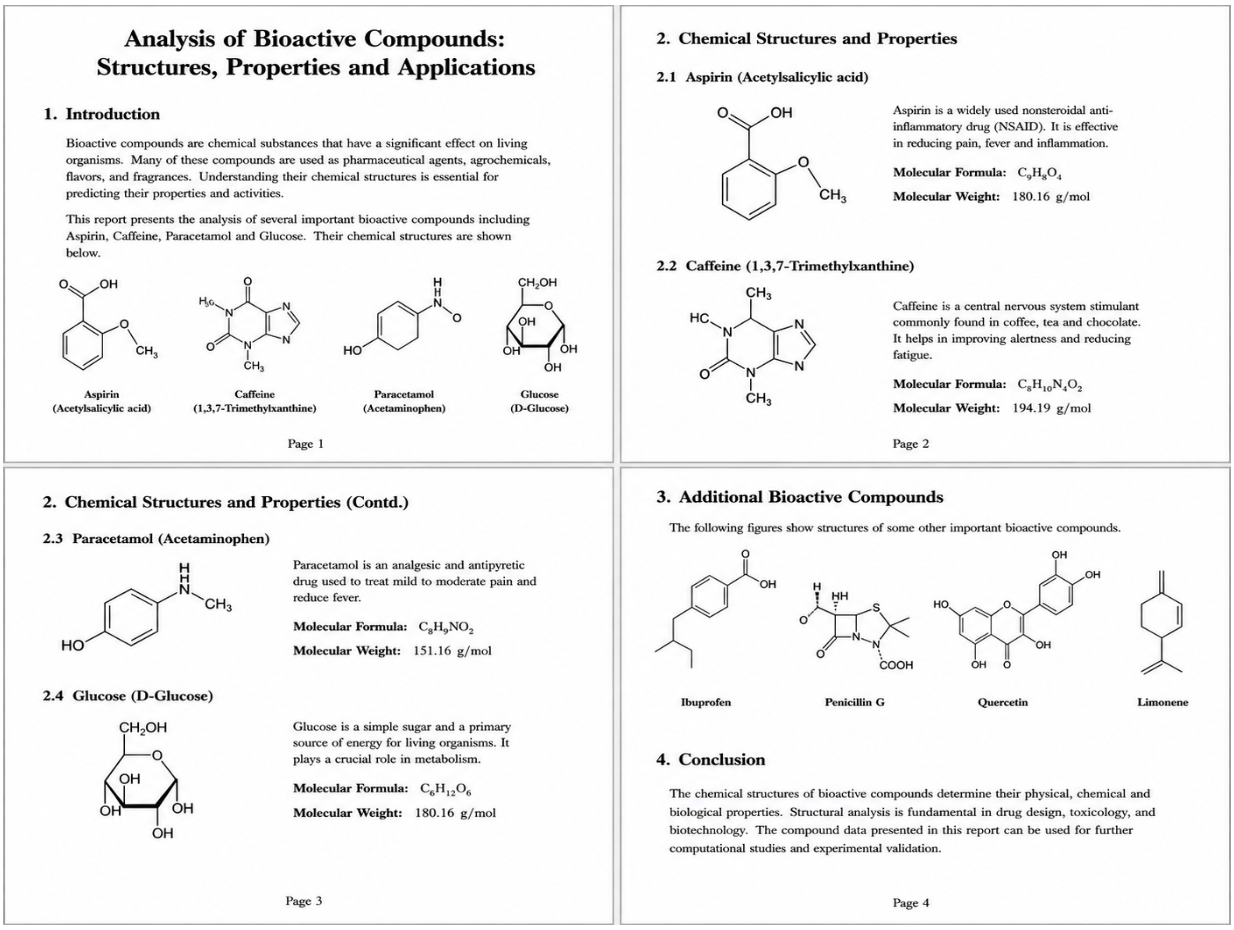
Page 1 - Organic Molecules Overview

This page introduces common molecular depictions used in scientific literature. Multiple chemical structures are shown below for OCR and segmentation testing.



Analysis of Bioactive Compounds: Structures, Properties and Applications

1. Introduction

Bioactive compounds are chemical substances that have a significant effect on living organisms. Many of these compounds are used as pharmaceutical agents, agrochemicals, flavors, and fragrances. Understanding their chemical structures is essential for predicting their properties and activities.

This report presents the analysis of several important bioactive compounds including Aspirin, Caffeine, Paracetamol and Glucose. Their chemical structures are shown below.

Aspirin (Acetylsalicylic acid) | Caffeine (1,3,7-Trimethylxanthine) | Paracetamol (Acetaminophen) | Glucose (D-Glucose)

Page 1

2. Chemical Structures and Properties

2.1 Aspirin (Acetylsalicylic acid)

Aspirin is a widely used nonsteroidal anti-inflammatory drug (NSAID). It is effective in reducing pain, fever and inflammation.

Molecular Formula: $C_9H_8O_4$

Molecular Weight: 180.16 g/mol

2.2 Caffeine (1,3,7-Trimethylxanthine)

Caffeine is a central nervous system stimulant commonly found in coffee, tea and chocolate. It helps in improving alertness and reducing fatigue.

Molecular Formula: $C_8H_{10}N_4O_2$

Molecular Weight: 194.19 g/mol

Page 2

2. Chemical Structures and Properties (Contd.)

2.3 Paracetamol (Acetaminophen)

Paracetamol is an analgesic and antipyretic drug used to treat mild to moderate pain and reduce fever.

Molecular Formula: $C_8H_9NO_2$

Molecular Weight: 151.16 g/mol

2.4 Glucose (D-Glucose)

Glucose is a simple sugar and a primary source of energy for living organisms. It plays a crucial role in metabolism.

Molecular Formula: $C_6H_{12}O_6$

Molecular Weight: 180.16 g/mol

Page 3

3. Additional Bioactive Compounds

The following figures show structures of some other important bioactive compounds.

Ibuprofen | Penicillin G | Quercetin | Limonene

4. Conclusion

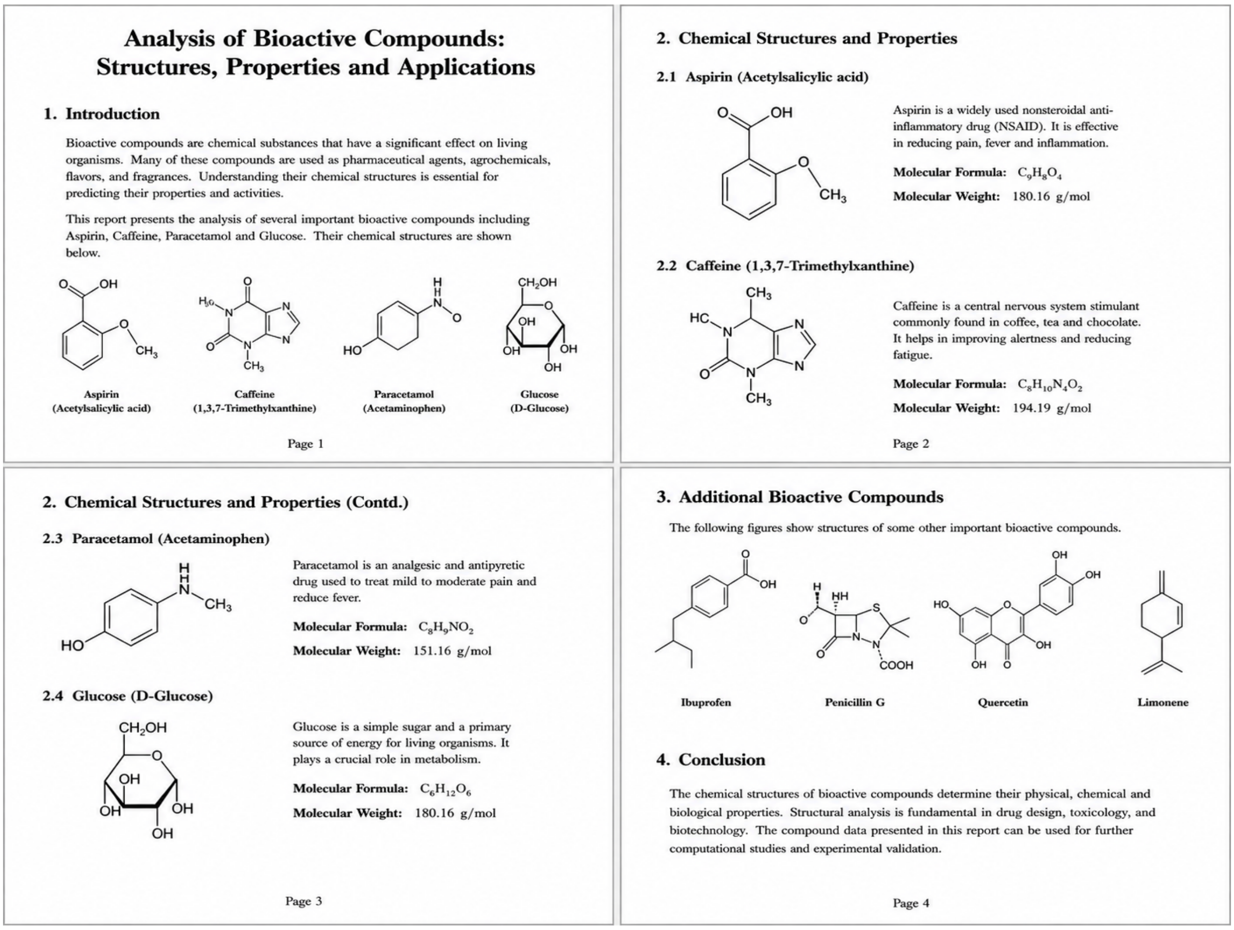
The chemical structures of bioactive compounds determine their physical, chemical and biological properties. Structural analysis is fundamental in drug design, toxicology, and biotechnology. The compound data presented in this report can be used for further computational studies and experimental validation.

Page 4

Figure: Example chemical structure plate for testing DECIMER segmentation and OCR.

Page 2 - Drug-like Compounds

This page simulates a medicinal chemistry paper containing several drug candidates and aromatic compounds.



Analysis of Bioactive Compounds: Structures, Properties and Applications

1. Introduction

Bioactive compounds are chemical substances that have a significant effect on living organisms. Many of these compounds are used as pharmaceutical agents, agrochemicals, flavors, and fragrances. Understanding their chemical structures is essential for predicting their properties and activities.

This report presents the analysis of several important bioactive compounds including Aspirin, Caffeine, Paracetamol and Glucose. Their chemical structures are shown below.

Aspirin (Acetylsalicylic acid) | **Caffeine** (1,3,7-Trimethylxanthine) | **Paracetamol** (Acetaminophen) | **Glucose** (D-Glucose)

Page 1

2. Chemical Structures and Properties

2.1 Aspirin (Acetylsalicylic acid)

Aspirin is a widely used nonsteroidal anti-inflammatory drug (NSAID). It is effective in reducing pain, fever and inflammation.

Molecular Formula: $C_9H_8O_4$

Molecular Weight: 180.16 g/mol

2.2 Caffeine (1,3,7-Trimethylxanthine)

Caffeine is a central nervous system stimulant commonly found in coffee, tea and chocolate. It helps in improving alertness and reducing fatigue.

Molecular Formula: $C_8H_{10}N_4O_2$

Molecular Weight: 194.19 g/mol

Page 2

2. Chemical Structures and Properties (Contd.)

2.3 Paracetamol (Acetaminophen)

Paracetamol is an analgesic and antipyretic drug used to treat mild to moderate pain and reduce fever.

Molecular Formula: $C_8H_9NO_2$

Molecular Weight: 151.16 g/mol

2.4 Glucose (D-Glucose)

Glucose is a simple sugar and a primary source of energy for living organisms. It plays a crucial role in metabolism.

Molecular Formula: $C_6H_{12}O_6$

Molecular Weight: 180.16 g/mol

Page 3

3. Additional Bioactive Compounds

The following figures show structures of some other important bioactive compounds.

Ibuprofen | **Penicillin G** | **Quercetin** | **Limonene**

4. Conclusion

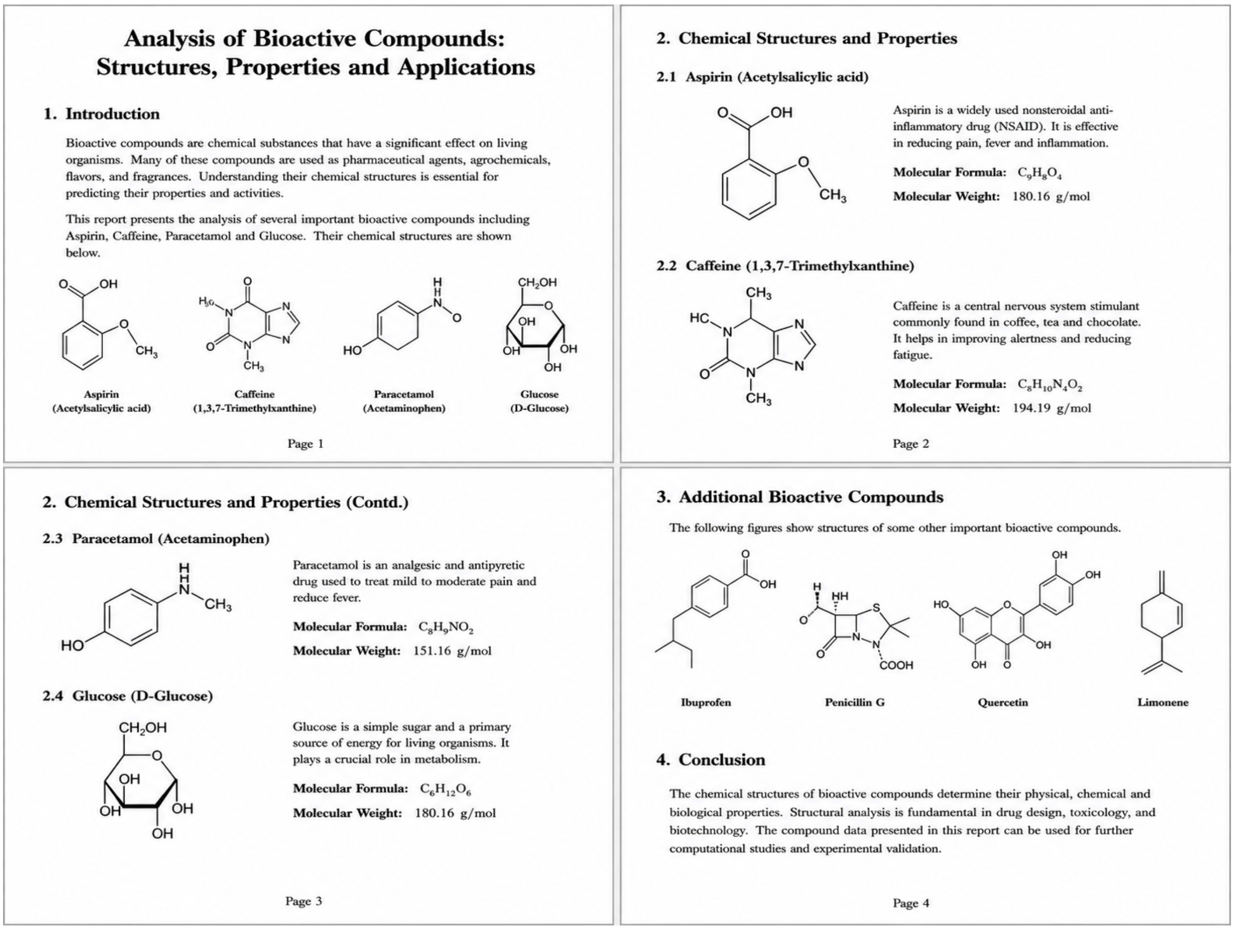
The chemical structures of bioactive compounds determine their physical, chemical and biological properties. Structural analysis is fundamental in drug design, toxicology, and biotechnology. The compound data presented in this report can be used for further computational studies and experimental validation.

Page 4

Figure: Example chemical structure plate for testing DECIMER segmentation and OCR.

Page 3 - Natural Products

This page mimics a journal article describing natural products and biochemical compounds. The embedded figures represent chemical structure diagrams.



Analysis of Bioactive Compounds: Structures, Properties and Applications

1. Introduction

Bioactive compounds are chemical substances that have a significant effect on living organisms. Many of these compounds are used as pharmaceutical agents, agrochemicals, flavors, and fragrances. Understanding their chemical structures is essential for predicting their properties and activities.

This report presents the analysis of several important bioactive compounds including Aspirin, Caffeine, Paracetamol and Glucose. Their chemical structures are shown below.

Aspirin (Acetylsalicylic acid) **Caffeine (1,3,7-Trimethylxanthine)** **Paracetamol (Acetaminophen)** **Glucose (D-Glucose)**

Page 1

2. Chemical Structures and Properties

2.1 Aspirin (Acetylsalicylic acid)

Aspirin is a widely used nonsteroidal anti-inflammatory drug (NSAID). It is effective in reducing pain, fever and inflammation.

Molecular Formula: $C_9H_8O_4$

Molecular Weight: 180.16 g/mol

2.2 Caffeine (1,3,7-Trimethylxanthine)

Caffeine is a central nervous system stimulant commonly found in coffee, tea and chocolate. It helps in improving alertness and reducing fatigue.

Molecular Formula: $C_8H_{10}N_4O_2$

Molecular Weight: 194.19 g/mol

Page 2

2. Chemical Structures and Properties (Contd.)

2.3 Paracetamol (Acetaminophen)

Paracetamol is an analgesic and antipyretic drug used to treat mild to moderate pain and reduce fever.

Molecular Formula: $C_8H_9NO_2$

Molecular Weight: 151.16 g/mol

2.4 Glucose (D-Glucose)

Glucose is a simple sugar and a primary source of energy for living organisms. It plays a crucial role in metabolism.

Molecular Formula: $C_6H_{12}O_6$

Molecular Weight: 180.16 g/mol

Page 3

3. Additional Bioactive Compounds

The following figures show structures of some other important bioactive compounds.

Ibuprofen **Penicillin G** **Quercetin** **Limonene**

4. Conclusion

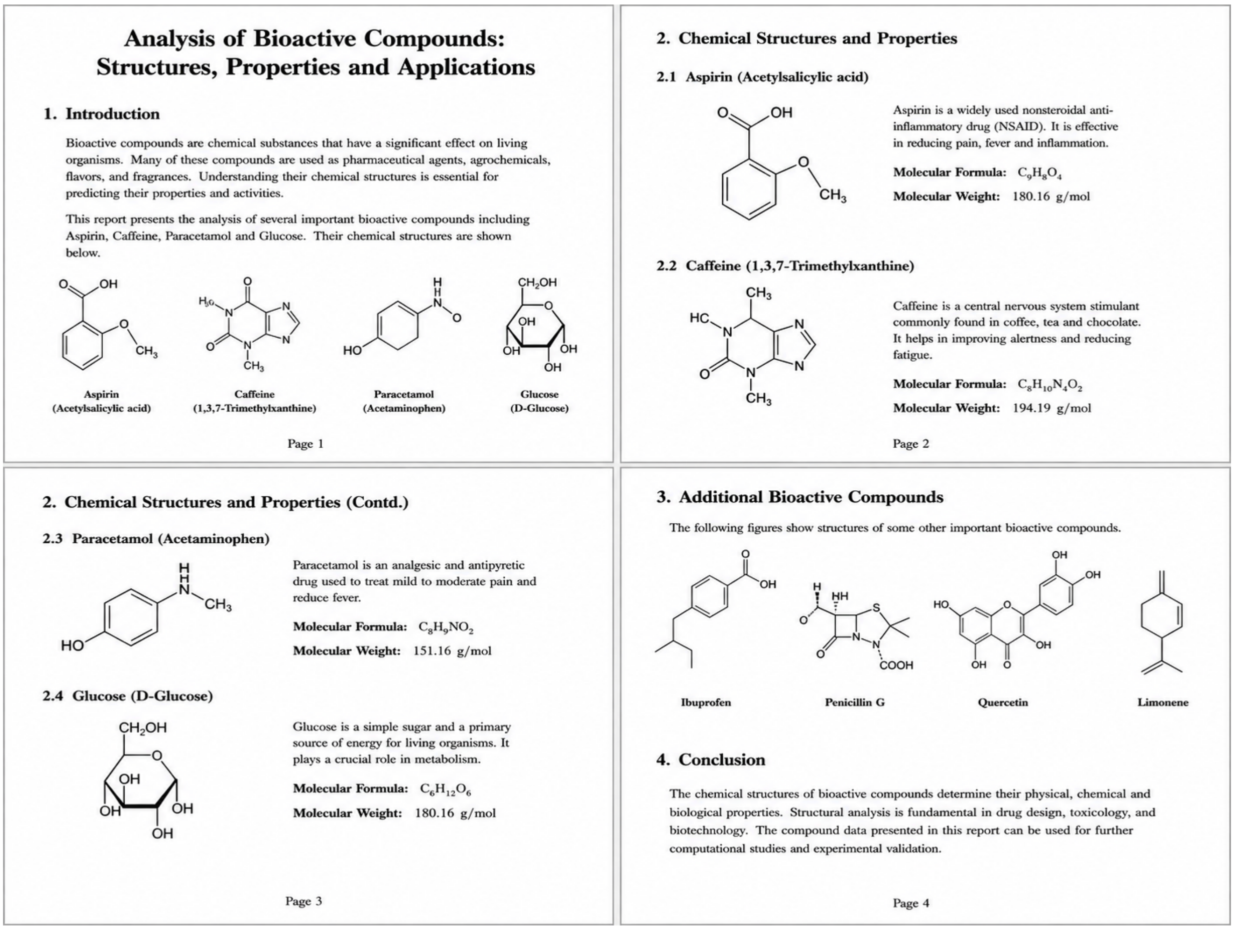
The chemical structures of bioactive compounds determine their physical, chemical and biological properties. Structural analysis is fundamental in drug design, toxicology, and biotechnology. The compound data presented in this report can be used for further computational studies and experimental validation.

Page 4

Figure: Example chemical structure plate for testing DECIMER segmentation and OCR.

Page 4 - Summary and Validation

This final page summarizes the document and contains additional figures for validating rendering, segmentation, and SMILES prediction.



Analysis of Bioactive Compounds: Structures, Properties and Applications

1. Introduction

Bioactive compounds are chemical substances that have a significant effect on living organisms. Many of these compounds are used as pharmaceutical agents, agrochemicals, flavors, and fragrances. Understanding their chemical structures is essential for predicting their properties and activities.

This report presents the analysis of several important bioactive compounds including Aspirin, Caffeine, Paracetamol and Glucose. Their chemical structures are shown below.

Aspirin (Acetylsalicylic acid) **Caffeine (1,3,7-Trimethylxanthine)** **Paracetamol (Acetaminophen)** **Glucose (D-Glucose)**

Page 1

2. Chemical Structures and Properties

2.1 Aspirin (Acetylsalicylic acid)

Aspirin is a widely used nonsteroidal anti-inflammatory drug (NSAID). It is effective in reducing pain, fever and inflammation.

Molecular Formula: $C_9H_8O_4$

Molecular Weight: 180.16 g/mol

2.2 Caffeine (1,3,7-Trimethylxanthine)

Caffeine is a central nervous system stimulant commonly found in coffee, tea and chocolate. It helps in improving alertness and reducing fatigue.

Molecular Formula: $C_8H_{10}N_4O_2$

Molecular Weight: 194.19 g/mol

Page 2

2. Chemical Structures and Properties (Contd.)

2.3 Paracetamol (Acetaminophen)

Paracetamol is an analgesic and antipyretic drug used to treat mild to moderate pain and reduce fever.

Molecular Formula: $C_8H_9NO_2$

Molecular Weight: 151.16 g/mol

2.4 Glucose (D-Glucose)

Glucose is a simple sugar and a primary source of energy for living organisms. It plays a crucial role in metabolism.

Molecular Formula: $C_6H_{12}O_6$

Molecular Weight: 180.16 g/mol

Page 3

3. Additional Bioactive Compounds

The following figures show structures of some other important bioactive compounds.

Ibuprofen **Penicillin G** **Quercetin** **Limonene**

4. Conclusion

The chemical structures of bioactive compounds determine their physical, chemical and biological properties. Structural analysis is fundamental in drug design, toxicology, and biotechnology. The compound data presented in this report can be used for further computational studies and experimental validation.

Page 4

Figure: Example chemical structure plate for testing DECIMER segmentation and OCR.